ADEL MAGDY MACKNOTA

Full Stack Software Engineer | Fintech, Payments & Compliance Systems | .NET · NestJS · React · Next.js

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PROFESSIONAL SUMMARY

Full Stack Software Engineer working across the .NET and Node.js ecosystems, focused on fintech, payments and compliance systems: software that has to be correct, auditable and safe to retry. I build ASP.NET Core and NestJS backends on SQL Server and PostgreSQL, and React, Next.js and Angular frontends. Current work spans an e-invoicing integration hub for ZATCA Phase 2, an automated payment reconciliation platform, an AI-driven SEO automation engine, and live, Arabic-first e-commerce storefronts where I delivered measurable performance and SEO gains.

I care about architecture as much as working code: Clean Architecture, SOLID, idempotency, auditability, secure secret handling and automated testing, so systems stay correct under real-world load and maintainable as they grow. I use AI development tools to move faster on research and refactoring, with every design decision and merged change under my own review.

TECHNICAL SKILLS

- **Languages:** C#, TypeScript, JavaScript, Python, SQL (T-SQL), HTML5, CSS3
- **Backend:** .NET 8 / ASP.NET Core Web API, NestJS, Node.js, Entity Framework Core, Prisma, LINQ, REST APIs, SignalR, Hangfire, Redis/BullMQ, JWT & ASP.NET Core Identity
- **Frontend:** React 18, Next.js, Angular 18, TypeScript, Vite, Tailwind CSS, TanStack Query, React Hook Form + Zod, RxJS, i18next (Arabic/RTL), Responsive Design
- **Databases & Caching:** Microsoft SQL Server, PostgreSQL, Redis
- **Architecture & Patterns:** Clean / Onion Architecture, Repository & Unit of Work, Specification Pattern, CQRS, SOLID, RBAC, Idempotency & Retry Design, Multi-tenant Architecture
- **Integrations:** ZATCA E-Invoicing (Phase 2), Stripe, Microsoft Graph, Google Search Console API, LLM APIs (Gemini, Groq, Mistral, OpenRouter), Webhooks
- **Web Performance & SEO:** Code splitting, Core Web Vitals (CLS), bundle budgets, canonical URLs, JSON-LD, slug-based routing
- **DevOps & Tools:** Docker, Git & GitHub, GitHub Actions, Nginx, Postman, Visual Studio, VS Code
- **Testing & Methods:** Unit Testing (Vitest), E2E Testing (Playwright), API Testing, Agile / Scrum, Code Review, AI-assisted development (Claude Code, ChatGPT, Gemini)

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer May 2026 – Present

Asus Card — Asas Al-Betaqat for Trading Est. · Cairo, Egypt (On-site)

- Building backend platforms end to end across two stacks (.NET/React and NestJS/Next.js), owning architecture, database design, background processing, integrations and automated testing. The three backend platforms below are in active development and not yet publicly live.

ZATCA E-Invoicing Integration Hub — ***Validated against ZATCA Sandbox & Simulation***

Stack: NestJS, Next.js, TypeScript, PostgreSQL, Prisma, Redis/BullMQ, Docker

- Designed an integration hub that abstracts the full ZATCA e-invoicing compliance workflow behind one API, so client systems integrate once instead of implementing the specification themselves.
- Implemented the complete onboarding lifecycle: taxpayer/EGS registration, CSR generation and certificate handling, Compliance CSID, the 6/6 compliance checks, and Production CSID issuance.
- Delivered B2B Clearance and B2C Reporting flows with ICV/PIH sequencing, evidence tracking, and outbound webhooks for asynchronous status updates.
- Enforced strict environment isolation across Sandbox, Simulation and Production, with safeguards that block cross-environment and unauthorized calls.
- Hardened the platform with RBAC, encrypted credential storage, idempotency keys, queue-based retries and full audit logging; covered it with automated SDK, API and browser end-to-end tests.

Multi-Brand E-Commerce Storefronts — Purple Card & Ish7an — ***Live*** · ***purple-card.com*** · ***ish7an.com***

Stack: React 18, TypeScript, Vite, Tailwind CSS, TanStack Query, React Hook Form + Zod, i18next, Playwright, Nginx

- Contributed to a multi-tenant e-commerce platform where one React codebase serves many branded, Arabic-first stores for digital gift cards and game top-ups.
- Rebuilt the Purple Card storefront against a new design (checkout, profile, notifications, dark mode and a mobile redesign) and built its loyalty system from scratch: customer points, tiers, Shop with Points, and the merchant-side loyalty control center.
- Built Ish7an's brand theming layer, splash screen, category navigation, product activation guide and mobile layout; reworked its buy-now flow, currency conversion and payment layout.
- Cut store home pages from 1.9 MB to 220 KB of JavaScript by isolating brand and template code, and eliminated mobile layout shift (CLS from up to 0.81 to ~0).

- Fixed cross-store SEO on a client-rendered app: migrated catalog URLs to slugs, preserved server-rendered titles, canonical URLs and JSON-LD for Google, auto-generated bilingual FAQs for every product page, and added CI checks that guard SEO routing.

Payment Reconciliation Platform — ***In active development***
Stack: .NET, EF Core, React, SQL Server, Hangfire, SignalR

- Designing and building an automated reconciliation platform to replace the manual comparison of bank settlements, payment-gateway data and merchant transactions.
- Automated ingestion of bank and payment-gateway settlement files, with background jobs backfilling missing partner data.
- Implemented a matching and exception-detection engine covering mismatches, missing transactions, refunds, duplicates, fees, commissions and net settlements.
- Delivered a real-time dashboard (SignalR) and a monthly results view backed by atomic snapshot publishing for consistent, reproducible reporting.

AI-Powered SEO Automation Platform — ***In active development***
Stack: NestJS, Next.js, TypeScript, Prisma/PostgreSQL, Python, Google Search Console API, multi-provider LLMs

- Engineering an SEO intelligence and automation system for a large e-commerce catalog, replacing fragmented manual optimization, stale SEO data and unsafe bulk updates.
- Designed a workflow that groups related products, detects SEO gaps and generates field-level AI recommendations grounded in Google Search Console and live product data.
- Implemented multi-provider LLM routing with fallback, plus a fact-validation and content-quality layer that guards against hallucinated output.
- Built controlled review → approval → apply pipelines with live verification and rollback, protected by idempotency, stale-data detection and permission checks.

.NET Developer Intern Oct 2025 – Dec 2025
Smart Up — Cairo, Egypt (Hybrid)

- Contributed to the full SDLC of enterprise web applications using ASP.NET Core, handling backend logic and SQL Server database design across 3+ modules.
- Developed and optimized RESTful APIs serving multiple frontend clients, following clean code and Agile standards.
- Collaborated with the frontend team to integrate REST APIs and resolve cross-stack issues through daily sync-ups and clear documentation.

PERSONAL PROJECTS

Store — E-Commerce RESTful API — ***github.com/Macknota/E-Commerce***
Stack: .NET 8, ASP.NET Core Web API, EF Core, SQL Server, Redis, AutoMapper, Docker, Stripe

- Architected the backend with Clean (Onion) Architecture across 4 layers to enforce separation of concerns and SOLID principles.
- Implemented the Specification Pattern with a Generic Repository to remove repeated query logic across the codebase.
- Cached basket data in Redis to cut read latency compared with direct database queries.
- Secured 20+ endpoints with JWT authentication and role-based authorization; integrated Stripe payments and Docker deployment.

Social Connections Platform
Stack: .NET 8, Angular 18, SignalR, SQL Server, EF Core, ASP.NET Core Identity

- Built a responsive Angular 18 SPA with a modular component architecture and RxJS, with 10+ reusable components.
- Implemented real-time messaging and live presence with SignalR, behind a secure REST API (15+ endpoints) with JWT authentication, likes/matches modelling and cloud photo storage.

EDUCATION & TRAINING

B.Sc. in Computer Science — Benha University, Faculty of Science June 2023
Grade: Very Good · Military Service: Completed (Dec 2024)

Back-End Diploma — Route Academy 2025
Intensive training in C#, SQL Server, Entity Framework Core, and enterprise-level REST API development.

LANGUAGES

Arabic — Native | English — Professional Working Proficiency (B2) | Spanish — Beginner